



about the author

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about the book

This book aims to connect coding newcomers with Python programming by focusing on problem-solving. The problem-solving approach will center around problem decomposition, in which we take large complex problems and break them down into more easily solvable subproblems. The subproblems will be encapsulated into functions, directly linking to how the reader will learn to use Python efficiently. However, we will begin by constructing games we all know and love because the rules are simple and straightforward. Card games such as War and BlackJack. We will later move into more ambiguous problems. The reader

doesn't need to have domain knowledge, but it will still be a challenging task for them.

Typical job roles for the primary reader are

- Everyday workers and professionals: Most likely in non-technical roles in careers such as finance, marketing, or healthcare, trying to pivot into data-centric careers.
- Professionals seeking advancement: They want to move up in their current company by adding basic coding to their skillset, thus making them more valuable employees.
- Non-computer science majors: They have domain knowledge in other fields and are trying to move into the Tech industry, but require a foundational understanding of programming.
- Hobbyists: They seek a self-paced route to acquire the skills needed for Do-it-Yourself projects. They are motivated by self-interest rather than career development.

minimally qualified reader

Prerequisites

- No prior programming experience is required.
- Basic computer literacy: The reader can browse the internet and manage files and folders.
- Logical thinking abilities: The reader can follow logical instructions or sequences, although they don't need prior programming logic skills.

Takeaways

After reading this book, the reader will be able to:

1. Critically solve computational problems by employing higher-order thinking.

2. Tackle practical, real-world challenges by breaking them into smaller, manageable computational problems solvable with Python.
3. Develop basic Python applications, starting with familiar games like War and BlackJack, moving towards more complex problem solving.
4. Understand and apply the basics of Python in Google Colab Notebooks, enhancing learning through an interactive development environment.
5. Interpret and resolve common programming errors and pitfalls, fostering a robust problem-solving mindset.
6. Leverage Python's built-in functions and libraries to add advanced features to applications.
7. Use data types, control structures, functions, and object-oriented programming concepts to create dynamic and efficient Python code.
8. Integrate APIs into applications to enhance functionality and user experience.

These takeaways are derived from the book's goals to make Python accessible to absolute beginners by focusing on problem-solving and computational thinking, preparing readers for a future in various technical and data-centric careers.

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PART 2: BUILDING AND REFINING YOUR PYTHON SKILLS

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PART 3: ADVANCED PYTHON TECHNIQUES AND GAME DEVELOPMENT

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